

Denis Bolonin

Senior Software Engineer, **5+ years** of experience

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UK full-time work permit

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PROFESSIONAL SUMMARY

Senior Software Engineer with **over 5 years** of commercial experience, working in teams ranging from small startups to large fintech companies, both remote and on-site. Built full-cycle development processes, from gathering requirements to establishing fully automated release pipelines with test coverage and CI/CD. Expert in developing high-performance backend systems and integrating various technologies.

Being Agile certified, I'm involved in projects from start to finish - defining tasks, working with analysts' team and even customers, developing solutions, and managing releases. My team lead granted me the chance to take on some leadership responsibilities, so I had a positive experience mentoring new colleagues who were just starting out on our team.

Actively engaged in the open-source community, I've contributed to projects like ClickHouse, a prominent open-source DBMS. I'm a highly collaborative person, and that's not just in open source. My team and I recently won first place at a hackathon at Sber.

WORK EXPERIENCE

Senior Software Engineer

08/2023 - Present, 2+ years

[digital.sfpe](#), a software development startup

- Developed from the ground up a comprehensive software solution tailored for fire protection engineering companies to ensure regulatory compliance. This solution automates critical processes such as contract management, data validation, format control, minimizes errors and proactively monitors deadlines. The implementation of this software significantly improved efficiency, enabling a **single person** to manage the workload that previously required **3-4 full-time employees**.

- Implemented autofill process for EMERCOM contract notifications, reducing the processing time per notification from 8-10 minutes to under 1.5-2 minutes (**over 80% speed up**).

- Engineered an end-to-end subscription billing system (order intake, payment document parsing, invoicing, and reconciliation), replacing an almost fully manual workflow and eliminating human error across the billing pipeline.

Python Backend Software Engineer

07/2022 - 07/2024, 2 years

[Sber](#), one of the biggest fintech companies

- Developed a utility for financial models for a high-value investment project, specifically valued at 2 billion USD, **accelerating** model calculations **by over 50%** through concurrent processing.

- Developed an automated data pipeline for depositary settlements, enabling fully automated data streams and eliminating manual intervention.

- Developed a team productivity dashboard for Scrum Masters, enabling data-driven decision-making and **increasing team throughput by 15%**.

C++ Backend Software Developer

10/2020 - 05/2022, 1 year 8 months

[ClickHouse](#), opensource DBMS

- Researched and implemented UDP-based replication protocols (UDT, μ TP/uTP, ENet) for ClickHouse, evaluating their performance and laying groundwork for future QUIC protocol studies.

- Integrated Rust into ClickHouse build system, enabling C++ and Rust interoperability. Implemented and integrated the BLAKE3 hashing function, achieving almost **100% performance increase** with a new hashing function compared to existing functions.

- Implemented YAML configuration support for ClickHouse, providing users with a more readable and convenient configuration format.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, C, Rust, SQL, Bash, R

Backend Development: Django, FastAPI, Flask, REST APIs, Gunicorn, Nginx, gRPC

Web Scraping & Data Extraction: BeautifulSoup, Selenium

Frontend Development: jQuery, React

Databases & Data Storage: PostgreSQL, SQLite, ClickHouse, Redis

Cloud & DevOps: Docker, Kubernetes, OpenShift, Jenkins, Helm, AWS, Istio, Nexus, CI/CD

Monitoring & Analytics: Prometheus, Grafana, Kibana

Message Queues & Streaming: Apache Kafka, RabbitMQ, ZooKeeper, Apache Spark

Development Tools: VS Code, PyCharm, Jupyter Notebook, Git, Conda, CMake, Linux, Bitbucket, GitHub, GitLab

Project Management: Agile, Scrum, Jira, Confluence, Team Leadership, Mentoring, Figma

EDUCATION

Master's degree in Computer Science. Queen Mary University of London, London, UK Software and Data Engineering stream	09/2024 - 09/2025
Bachelor's degree in Computer Science. HSE University, Moscow, Russia Applied Mathematics and Informatics, Distributed systems	09/2019 - 09/2023

CERTIFICATIONS

ICAgile Certified Professional Completed Agile training, received the ICAgile Certified Professional Certificate Link	12/2022
Applied statistical analysis Completed additional Applied Statistical Analysis course, HSE University	05/2022

PUBLICATIONS

More Than 2x Faster Hashing in ClickHouse Using Rust
Blog post describing the task of Rust integration into ClickHouse, its results, and efficiency improvements.
ClickHouse CTO acknowledged the significant impact of this work in his comment in the post.
[ClickHouse Official Blog](#)